

1. Administrative & Study Identification

Full Study Title: A Phase III, Open-Label, Randomized Study to Investigate the Efficacy and Safety of Atezolizumab (Anti-PD-L1 Antibody) Compared With Best Supportive Care Following Adjuvant Cisplatin-Based Chemotherapy in Patients With Completely Resected Stage IB-IIIA Non-Small Cell Lung Cancer **Brief Study Title:** Study to Assess Safety and Efficacy of Atezolizumab (MPDL3280A) Compared to Best Supportive Care Following Chemotherapy in Patients With Lung Cancer [IMpower010] **Short Title/Acronym:** IMpower010 **Posting ID:** 444890969734721163 **Protocol Number:** G029527 **NCT Number:** NCT02486718 **Sponsor Name / Company:** Roche (F. Hoffmann-La Roche Ltd / Genentech, Inc.) **Investigational Product:** Atezolizumab (MPDL3280A / Tecentriq) **Phase of Development:** Phase III (PHASE3) **Trial Status:** Active but not currently recruiting (ACTIVE_NOT_RECRUITING) **Medical Monitor:** F. Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of 16 cycles of atezolizumab treatment compared with best supportive care (BSC) as measured by investigator-assessed disease-free survival (DFS). **Secondary Objectives:** To evaluate overall survival (OS) in the intention-to-treat (ITT) population, assess disease-free rates at year 3, and to assess the safety and tolerability of atezolizumab in the adjuvant setting.

2.2 Background & Rationale Non-small cell lung cancer (NSCLC) is the most common type of lung cancer, and patients have a poor prognosis with high recurrence rates (30%-70%) after surgical resection. Adjuvant chemotherapy is the standard of care in this setting, but a significant unmet need remains for novel therapies that improve survival outcomes. Atezolizumab blocks the programmed death-ligand 1 (PD-L1) pathway, which helps to reverse the suppression of the immune system and allows it to fight cancer. The IMpower010 trial aims to demonstrate that adjuvant immunotherapy following platinum-based chemotherapy improves DFS and OS compared to BSC.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator / Best Supportive Care (BSC) **Blinding:** Open-label **Randomization:** 1:1

3.2 Planned Enrollment & Duration Targeted Enrollment: 1,280 randomized participants **Number of Sites:** Global, multicenter **Estimated Study Start:** October 31, 2015 **Estimated Primary Completion:** January 26, 2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Complete tumor resection of Stage IB (tumors ≥ 4 cm) to Stage IIIA NSCLC (per UICC/AJCC staging system, 7th edition). 2. Completion of up to 4 cycles of adjuvant cisplatin-based doublet chemotherapy (e.g., cisplatin plus vinorelbine, docetaxel, gemcitabine, or pemetrexed). *Note on Chemotherapy: Drug Name: cisplatin (Preferred Name: cisplatin); Trade Name: Platinol; ATC Code: L01XA01.* 3. Evaluation of PD-L1 expression status.

4.2 Exclusion Criteria 1. Signs or symptoms of infection within 14 days prior to randomization. 2. Major surgical procedure within 28 days prior to randomization. 3. Treatment with systemic immunostimulatory agents or immunosuppressive medications (e.g., systemic corticosteroids) within specified pre-randomization timeframes.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Atezolizumab 1200 mg every 3 weeks (Q3W). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** 16 cycles (each cycle = 21 days), equating to approximately 1 year of treatment.

5.2 Concomitant & Prohibited Therapy Permitted: Best supportive care (in the control arm) and standard supportive medications. **Prohibited:** Systemic corticosteroids, immunosuppressive medications, and live attenuated vaccines during the treatment period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Pre-enrollment Informed Consent, Medical History, Tumor resection, PD-L1 evaluation Enrollment Phase
Up to 4 cycles	Cisplatin-based (Platinol) adjuvant chemotherapy administration	Baseline	Day 0 Randomization (1:1), First Dose of Atezolizumab or initiation of BSC
Interim	Every 3 weeks	IV Atezolizumab administration, Safety Labs, periodic chest X-ray and CT scan	End of Study / Follow-up
Up to 5+ years	Disease-free survival and Overall survival long-term follow-up		

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Disease-free survival (DFS) as assessed by the investigator, evaluated hierarchically in the Stage II-III A PD-L1 tumor cell (TC) $\geq 1\%$ population, all-randomized Stage II-III A population, and finally the ITT (Stage IB-III A) population. **Measurement Method:** Periodic radiographic assessments (CT/MRI) and clinical evaluation.

7.2 Secondary & Exploratory Endpoints 1. Overall Survival (OS) in the ITT Population. 2. Disease-free Rate at Year 3 in ITT Population. 3. Disease-free Rate at Year 3 in All Randomized Stage II-III A Population. 4. DFS by KRAS mutational status (exploratory). 5. Incidence and severity of adverse events (Safety and tolerability).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard laboratory testing and clinical evaluation during Q3W visits. Common AEs include increased AST/ALT, hyperkalemia, rash, hypothyroidism, pyrexia, fatigue, and peripheral neuropathy. **Serious Adverse Events (SAEs):** Standard 24-hour reporting protocols for severe immune-mediated adverse events (imAEs). **Data Monitoring Committee (DMC):** External independent data monitoring committee to periodically review safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) population, Stage II-III A all-randomized, and Stage II-III A PD-L1 TC $\geq 1\%$. **Sample Size Justification:** Designed to appropriately power sequential hierarchical testing of DFS and OS across the target PD-L1 subgroups and broad ITT populations up to the target enrollment of 1,280 participants. **Handling of Missing Data:** Standard Kaplan-Meier methodologies and stratified hazard ratios (HR) with 95% Confidence Intervals for survival analysis.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Centralized and onsite monitoring of global sites; strict adherence tracking to infusion schedules. **Audit Trail:** eCRF locking and comprehensive documentation of radiographic scans for independent review.

11. Study Identifiers & Governance

Primary ID: G029527 **Posting ID:** 444890969734721163 **Registry Number:** NCT02486718 **Sponsor/Company:** Roche (F. Hoffmann-La Roche Ltd / Genentech) **Study Title:** IMpower010 **Official Regulatory Title:** A Phase III, Open-Label, Randomized Study to Investigate the Efficacy and Safety of Atezolizumab Compared With Best Supportive Care Following Adjuvant Cisplatin-Based Chemotherapy in Patients With Completely Resected Stage IB-IIIA Non-Small Cell Lung Cancer **Trial Status:** ACTIVE_NOT_RECRUITING

12. Clinical Framework (NSCLC Specific)

Disease Areas: Non-Small Cell Lung Cancer **Primary Condition:** Stage IB-IIIA Non-Small Cell Lung Cancer (NSCLC) **Preferred UMLS Name:** Non-Small Cell Lung Carcinoma **Related Conditions:** Metastatic Non Small Cell Lung Cancer (Preferred Name: Metastatic non-small cell lung cancer) **Diagnosis Threshold:** Completely resected tumor and completed adjuvant platinum-based chemotherapy **Medical Methodology:** Adjuvant Immunotherapy (PD-L1 Inhibition) **Optimization Target:** Prolongation of Disease-Free Survival (DFS) and Overall Survival (OS)

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for adjuvant NSCLC treatment **Education Protocol (HCP):** Site initiation visits and immune-mediated adverse event (imAE) management training **Education Protocol (Patient):** Infusion education and AE self-monitoring **Quality Audit Mechanism:** Centralized site monitoring, radiological scan central review, and eCRF queries

14. Observation & Follow-Up Schedule

Total Duration: ≥ 5 years (60+ months) of survival follow-up **Onsite Visit Cadence:** Every 3 weeks (Q3W) during the 16-cycle treatment phase **Remote Monitoring Frequency:** Routine survival follow-up post-treatment **Checklist Review Interval:** Periodic radiological imaging (CT/chest X-ray) intervals per protocol

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				